# A IOT-DRIVEN SMART HOSTEL MANAGEMENT AND STUDENT PROTECTION ECOSYSTEM

## A PROJECT REPORT

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*in partial fulfillment for the award of the degree*

*of*

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**OCTOBER 2026**

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# ABSTRACT

A IoT-Driven Smart Hostel Management and Student Protection Ecosystem is an integrated smart hostel solution designed to improve student safety, attendance, resource monitoring, communication, and complaint management. The proposed system combines Artificial Intelligence, Internet of Things, embedded systems, mobile applications, and real-time monitoring to create a secure and efficient hostel environment.

A fingerprint-based attendance system automatically records student entry and exit. A smart safety robot equipped with a camera and sensors detects potential hazards such as snakes and generates immediate alerts. A water quality monitoring module continuously monitors water conditions and provides alerts when abnormal parameters are detected. A parent notification system enables important student-related and emergency information to be communicated to parents.

In addition, a dedicated hostel complaint management application allows students to register complaints related to maintenance, electrical issues, water supply, cleanliness, and other hostel facilities. Hostel administrators can view, manage, and update the status of complaints, improving transparency and response time. By integrating these modules into a single a IoT ecosystem, the proposed system reduces manual monitoring, improves communication between students and management, enables faster emergency response, and provides a centralized platform for smart hostel management and student protection.

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# LIST OF ABBREVIATIONS

| ABBREVATION | DEFINITION |
|---|---|
| AI | Artificial Intelligence |
| AIoT | Artificial Intelligence of Things |
| IoT | Internet of Things |
| ESP32 | Espressif Systems 32-bit Microcontroller |
| ESP32-CAM | ESP32 Camera Module |
| PIR | Passive Infrared |
| LDR | Light Dependent Resistor |
| RTC | Real-Time Clock |
| SMS | Short Message Service |
| GPIO | General-Purpose Input/Output |
| Wi-Fi | Wireless Networking Technology |
| USB | Universal Serial Bus |
| IDE | Integrated Development Environment |
| YOLO | You Only Look Once |
| GPS | Global Positioning System |
| UI | User Interface |
| DB | Database |
| API | Application Programming Interface |
| HTTP | Hypertext Transfer Protocol |
| HTTPS | Hypertext Transfer Protocol Secure |
| GND | Ground |
| VCC | Supply Voltage |
| LED | Light-Emitting Diode |
| DC | Direct Current |
| AC | Alternating Current |
| PCB | Printed Circuit Board |
| UART | Universal Asynchronous Receiver-Transmitter |
| Blynk | IoT Monitoring and Control Platform |

# CHAPTER – 1

# INTRODUCTION

## INTRODUCTION

The increasing student population in hostels has created a need for efficient management, improved security, automated attendance monitoring, and reliable communication. Traditional hostel management methods primarily depend on manual processes, which may lead to administrative difficulties, delayed communication, and inadequate safety monitoring. Integrating Artificial Intelligence (AI), the Internet of Things (IoT), embedded systems, and mobile applications provide an effective approach to addressing these challenges.

The project **"A IoT-Driven Smart Hostel Management and Student Protection Ecosystem"** integrates five major modules: fingerprint-based attendance, smart hostel monitoring, AI-based safety robot, parent notification, and hostel complaint management application. The system uses biometric identification to record attendance, sensors such as PIR and LDR for movement detection and automated lighting control, and an ESP32-CAM-based robot for potential hazard detection, particularly snakes. IoT-based notifications facilitate timely communication with parents and hostel administrators, while the complaint management application enables students to submit complaints and track their resolution.

The primary objective of this project is to develop a smart, secure, and efficient hostel management ecosystem that enhances student protection, reduces manual intervention, automates routine operations, and improves communication among students, parents, and hostel authorities. By integrating AI, IoT, and embedded technologies, the proposed system aims to provide a safer residential environment and improve overall hostel administration.

## 1.1OBJECTIVES

The primary objectives of the project are as follows:

- To implement an automated fingerprint-based attendance system for accurate student identification and systematic recording of entry and exit information.

- To develop an IoT-enabled hostel monitoring system using PIR, LDR, and RTC modules to monitor movement, ambient lighting conditions, and time-dependent operations.
- To automate hostel lighting based on environmental conditions and predefined schedules using relay-based control mechanisms.
- To design an AI-based safety robot incorporating ESP32-CAM for detecting potential hazards, particularly snakes, and generating timely alerts.
- To integrate the Blynk IoT platform for remote monitoring of relevant hostel activities and the operational status of connected devices.
- To improve student safety, streamline hostel administration, minimize manual intervention, and enhance operational efficiency through the integration of AI, IoT, and embedded technologies.

## 1.2 EXISTING SYSTEM

- Conventional hostel management systems primarily rely on manual procedures and independent mechanisms for student attendance, security monitoring, complaint handling, and parent communication. Student attendance is generally maintained through registers or basic recording systems, while hostel premises and student activities are monitored through periodic physical supervision by wardens and administrative staff.

- In most traditional systems, students are required to report maintenance and facility-related issues directly to hostel authorities. Communication with parents regarding attendance, student movement, and emergency situations may also be delayed due to the absence of automated notification mechanisms. Furthermore, safety monitoring depends largely on human observation, which may limit the ability to identify potential hazards and respond promptly to unexpected incidents. The major limitations of the existing system are as follows:

- Manual attendance recording may result in inaccuracies and difficulties in maintaining student entry and exit records.

- Continuous human supervision is required to monitor hostel activities and ensure student safety.

- The absence of integrated monitoring limits real-time visibility into hostel environmental conditions and device status.

- Limited remote accessibility restricts real-time monitoring and timely decision-making by hostel administrators.

## 1.3 PROPOSED SYSTEM

The proposed A IoT-Driven Smart Hostel Management and Student Protection Ecosystem integrates AI, IoT, embedded systems, and mobile applications to improve hostel automation, student safety, and administrative efficiency.

The system consists of five major modules: fingerprint-based attendance management, smart hostel monitoring, an AI-based safety robot, parent notification, and a hostel complaint management application. Sensors such as PIR and LDR, along with an RTC and relay, support movement monitoring, time-based operations, and automatic lighting control.

An ESP32-CAM-based robot assists in detecting potential hazards, such as snakes, and generating alerts. Parent notifications provide relevant student entry/exit and safety updates, while the complaint application enables students to submit complaints and track their status as Pending, In Progress, or Resolved. The Blynk IoT platform and Wi-Fi connectivity enable remote monitoring and communication between compatible modules.

The proposed system aims to provide a smart, secure, efficient, and scalable hostel environment through automated monitoring, timely notifications, and digital complaint management.

## 1.4 ADVANTAGES

- **Automated Attendance Management:** Fingerprint-based identification automates attendance recording and reduces manual effort and recording errors.

- **Enhanced Student Safety:** The ESP32-CAM-based safety robot supports the identification of potential hazards and facilitates timely alert generation.

- **Real-Time Monitoring:** Sensor integration and the Blynk IoT platform enable remote monitoring of connected devices and relevant hostel parameters.

- **Intelligent Lighting Control:** Automated lighting based on ambient light conditions and predefined schedules improves energy efficiency.

- **Timely Parent Notifications:** IoT-based communication facilitates the delivery

of important attendance, entry/exit, and safety-related alerts.

- **Efficient Complaint Management:** The digital application simplifies complaint registration, status tracking, and maintenance coordination.
- **Reduced Manual Intervention:** Integration of multiple hostel functions minimizes dependence on repetitive manual procedures.
- **Centralized Administration:** The combined system improves coordination among students, parents, and hostel authorities.
- **Scalability and Flexibility:** The modular architecture supports future expansion with additional sensors, monitoring functions, and communication services.

## 1.5 APPLICATIONS

- **College and University Hostels**: Used for student attendance, safety monitoring, and hostel management.
- **Student Attendance Management:** Used to record student attendance through fingerprint authentication.
- **Hostel Safety and Security:** Used to identify potential hazards using an ESP32-CAM-based safety robot.
- **Parent Notification System**: Used to send SMS alerts to parents through the SIM800L GSM module.
- **Smart Hostel Automation**: Used for automatic lighting control using PIR, LDR, RTC, and relay modules.
- **Hostel Complaint Management**: Used to submit complaints and track their status through a mobile application.
- **Smart Campus Management**: Used for centralized monitoring and improved communication among students, parents, and hostel administrators.

## 1.6 LITERATURE SURVEY

**1.1 LITERATURE SURVEY**

| S. No. | Author / Year | Methodology / System Description | Technology Used | Advantages | Limitations |
|---|---|---|---|---|---|
| 1 | Kumar et al., 2022 | Developed an IoT-based smart hostel monitoring system for monitoring student activities and hostel conditions. | IoT, ESP32, Sensors, Wi-Fi | Real-time monitoring and reduced manual supervision. | Requires Internet connectivity. |
| 2 | Patel et al., 2021 | Designed a fingerprint-based attendance system for automatic student identification and attendance recording. | Fingerprint Sensor, Arduino, Database | Accurate identification and reduced attendance errors. | Limited to registered users and biometric dependency. |
| 3 | Sharma et al., 2022 | Proposed an IoT-based smart home/hostel automation system for automatic lighting and appliance control. | ESP32, PIR, LDR, Relay, IoT | Saves energy and provides automatic control. | Sensor errors may affect automation. |
| 4 | Rahman et al., 2023 | Developed a surveillance robot using a camera and wireless communication for monitoring and safety applications. | ESP32-CAM, Wi-Fi, Robot, Camera | Remote monitoring and quick response to hazards. | Camera performance depends on lighting and network quality. |
| 5 | Singh et al., 2023 | Proposed a mobile-based complaint management system for students to submit and track hostel-related complaints. | Mobile App, Database, Cloud | Easy complaint registration and status tracking. | Requires smartphone and Internet access. |

# CHAPTER 2

# SYSTEM ARCHITECTURE AND COMPONENTS

## 2.1 COMPONENTS

This chapter details the foundational hardware and software elements selected for their optimal performance and synergistic integration within A IoT-Driven Smart Hostel Management and Student Protection Ecosystem.

### 2.1.1 CORE PROCESSING UNIT: ESP32 MICROCONTROLLER

The ESP32 is the main microcontroller used in the IoT-Driven Hostel Management and Student Protection Ecosystem. It receives data from sensors such as the PIR sensor and fingerprint module to monitor room occupancy and student attendance. It processes the collected data and performs the required control operations. Using its built-in Wi-Fi, the ESP32 connects to the Blynk IoT platform for real-time monitoring and status updates. It can also interface with a GSM module to send SMS alerts to parents regarding student attendance and safety. The RTC module enables time-based monitoring and automated operations during specified intervals. The ESP32 integrates various hostel management and student protection modules into a unified automated system. Its compact size, processing capability, and wireless connectivity make it suitable for smart hostel applications. Overall, the ESP32 enhances automation, monitoring efficiency, and student safety.




**Fig 2.1 ESP32-DEV KIT**

## 2.1.2 ESP32-CAM

The ESP32-CAM is the low-cost, compact development board that combines an ESP32 microcontroller with a camera module, commonly the OV2640. It supports 2.4 GHz Wi-Fi, enabling it to capture images and stream live video to a mobile phone or computer. It also supports MicroSD card storage for saving images and videos. The ESP32-CAM can be programmed using the Arduino IDE and is used in various applications, such as home security systems, face recognition, attendance monitoring, smart door locks, and IoT surveillance projects. Since the common ESP32-CAM board does not have a built-in USB programming connector, a USB-to-TTL adapter is generally required to upload programs. Its compact size, wireless connectivity, and camera capabilities make it a useful choice for smart monitoring and automation projects.




**Fig 2.2 ESP32-CAM**

## 2.1.3 R307S FINGERPRINT SENSOR

The fingerprint sensor is used for biometric student identification in the Smart Attendance module. Each student's fingerprint can be registered in the sensor before using the system. During attendance, the sensor scans and compares the student's fingerprint. The identified fingerprint information is processed by the controller. A successful match can be used to mark the student's attendance. It provides an additional identification method along with RFID. Fingerprint authentication helps reduce manual attendance work. It provides quick and convenient biometric identification. The sensor can communicate with the ESP32 for processing the attendance data. Thus, it forms an important part of the multi-mode smart attendance system.



**Fig 2.3 R307S Fingerprint Sensor**

## 2.1.4 HC-SR04 ULTRASONIC SENSOR

The HC-SR04 ultrasonic sensor is used to monitor the waste level in the Smart Dustbin module. It is placed at the top of the dustbin facing the waste. The sensor sends ultrasonic waves and receives the reflected echo. The ESP32 calculates the distance between the sensor and the waste surface. A smaller distance indicates that the dustbin is becoming full. The system compares the measured distance with predefined threshold values. When the dustbin is full, the red LED and buzzer can be activated. The dustbin status can also be displayed on the LCD. The status can be transmitted to the Blynk IoT platform for remote monitoring.




**Fig 2.4 HC-SR04 Ultrasonic Sensor**

## 2.1.5 PIR MOTION SENSOR

The PIR sensor is used to detect human movement inside the classroom. It detects changes in infrared radiation caused by a moving person. The sensor sends a signal to the ESP32 when motion is detected. The controller uses this information to determine whether the classroom is occupied. It works together with the LDR sensor in the automatic lighting module. When movement is detected and the classroom is dark, the system can turn the light ON. When no movement is detected, the system can turn the light OFF. This helps prevent unnecessary usage of classroom lighting. The PIR

sensor provides simple and automatic occupancy detection. Thus, it plays an important role in the Smart Light Automation module.



**Fig 2.5 Pir Motion Sensor**

## 2.1.6 MOTOR DRIVER

Motor Driver (L298N) The motor driver is used to control the speed and direction of the DC motor. The L298N motor driver acts as an interface between the microcontroller and the motor. It receives control signals from the microcontroller and provides the required power to operate the motor. The motor driver allows the motor to rotate in both clockwise and anticlockwise directions by controlling the input signals. It also supports speed control using Pulse Width Modulation (PWM). The motor driver is widely used in robotics, automation, and IoT-based applications. It helps control motor operations efficiently and enables the automation of system.

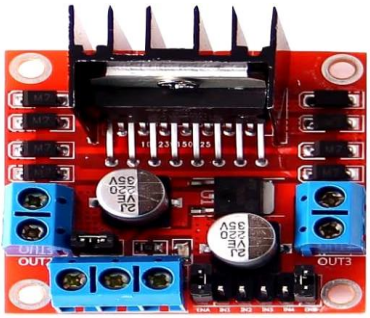


**Fig 2.6 Motor Driver**

## 2.1.7 RTC DS 1307

RTC DS1307 (Real-Time Clock Module) The DS1307 Real-Time Clock (RTC) module is an electronic component used to maintain accurate date and time information in embedded systems. It communicates with microcontrollers through the I²C communication protocol and uses a backup battery to retain timekeeping information

even when the main power supply is disconnected. The module provides essential time-related data, including seconds, minutes, hours, days, dates, months, and years. In the proposed system, the RTC DS1307 is used to record the accurate date and time of student attendance and hostel entry and exit a ctivities. It helps maintain reliable timestamps for attendance records and parent notifications, improving the accuracy and efficiency of the automated hostel monitoring system.

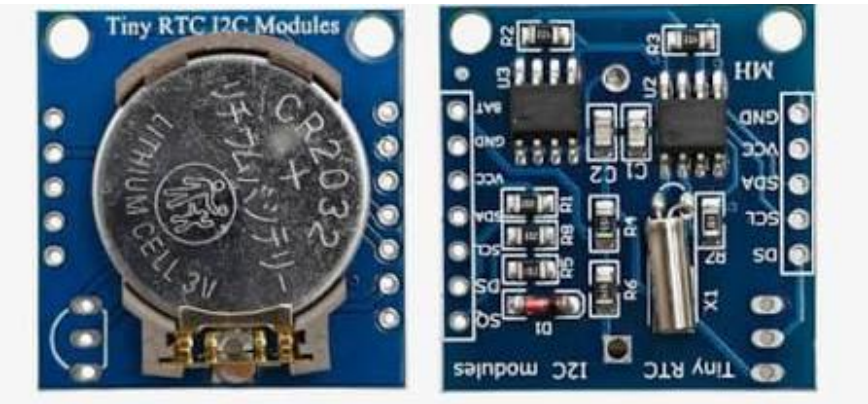



**Fig 2.7 RTC DS 1307**

## 2.1.8 GEAR MOTOR

A gear motor is an electromechanical device that combines an electric motor with a gearbox to produce rotational motion at a reduced speed and increased torque. The motor converts electrical energy into mechanical energy, while the gearbox uses gears to reduce the motor's speed and increase its turning force. Gear motors are available in different types, such as DC gear motors, AC gear motors, and stepper gear motors. They are commonly used in robotics, line-following robots, conveyor belts, electric vehicles, and industrial automation systems. In a robot, a DC gear motor helps the wheels rotate with sufficient torque to move the robot forward, backward, or turn. Gear motors are widely preferred because they provide controlled movement, high torque, and efficient operation in compact applications.

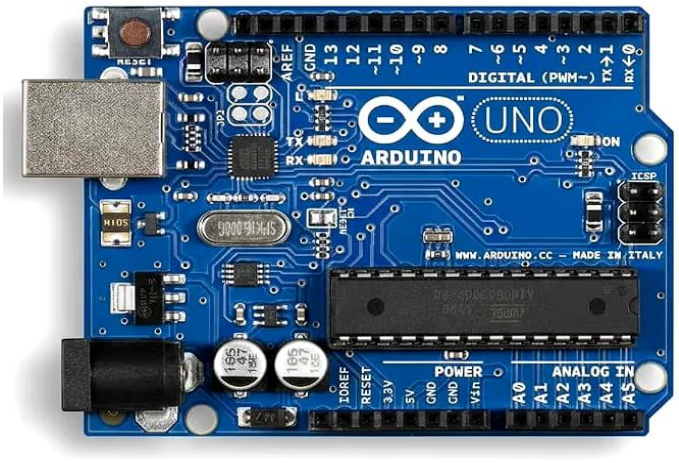


**Fig 2.8 Gear Motor**

## 2.1.9 ARDUINO UNO

The Arduino UNO is a microcontroller development board based on the ATmega328P microcontroller, widely used in embedded systems and automation applications. It provides digital and analog input/output pins for interfacing with various electronic components, sensors, and communication modules. In the proposed system, the Arduino UNO acts as a control unit that processes input signals and executes programmed instructions to coordinate the operation of connected hardware components. Its ability to support real-time control, sensor interfacing, and reliable hardware communication makes it suitable for developing automated student monitoring and attendance management systems.

**Fig 2.9 Arduino UNO**



## 2.1.10 BUZZER

The buzzer is used to provide an audible alert when an important condition is detected. It is controlled by the ESP32 according to the programmed conditions. In the Smart Dustbin module, the buzzer can indicate that the dustbin has reached its full level. In the student monitoring module, it can provide an alert when drowsiness is detected. The buzzer provides immediate notification without requiring the user to continuously observe the LCD. It can work together with red and green LEDs for clear status indication. The buzzer is simple to connect and control using an ESP32 GPIO pin. It provides a low-cost and effective alert mechanism. Its sound helps users quickly identify an abnormal or alert condition. Thus, the buzzer is an important notification component of our Smart Classroom Management System.



**Fig 2.10 Buzzer**

## 2.1.11 SERVO MOTOR

A servo motor is an electromechanical device used to control the angular position, speed, and movement of a shaft with high precision. It consists of a motor, a gear mechanism, a control circuit, and a feedback system. The servo motor receives a control signal, usually a PWM (Pulse Width Modulation) signal, to rotate its shaft to a desired angle, commonly between 0° and 180° for a standard positional servo. The feedback system helps the motor maintain the required position accurately. Servo motors are widely used in robotics, robotic arms, automatic doors, camera positioning systems, and industrial automation. They are commonly controlled using microcontrollers such as Arduino and ESP32, making them useful for applications that require precise position control.




**Fig 2.11 Servo Motor**

## 2.1.12 WHEELS

Wheels are circular mechanical components used to enable a robot or vehicle to move from one place to another. In robotics, wheels are usually connected to motors that provide rotational motion, allowing the robot to move forward, backward, and turn. They are commonly made of rubber or plastic to provide grip, stability, and smooth movement on different surfaces. Different types of wheels, such as standard wheels, caster wheels, omni wheels, and magnum wheels, are used depending on the robot's movement requirements. Wheels are widely used in line-following robots, obstacle-avoiding robots, mobile robots, and automated vehicles.

**Fig2.12 Wheels**



## 2.1.13 JUMPER WIRES

Jumper wires are electrical wires used to connect electronic components and modules without soldering. They are commonly used in breadboard circuits and microcontroller projects involving Arduino, ESP32, sensors, motors, and other electronic devices. Jumper wires are available in three main types: male-to-male, male-

to-female, and female-to-female, depending on the type of connectors at their ends. They are used to establish connections between power supply pins, ground pins, input/output pins, and other circuit components. Jumper wires are reusable, easy to connect, and useful for testing and prototyping electronic circuits.

**Fig 2.13 Jumper Wires**



## 2.1.14 LITHIUM- ION CELL

A lithium-ion cell is a rechargeable battery that stores electrical energy through the movement of lithium ions between its positive and negative electrodes. It consists of a cathode, an anode, an electrolyte, and a separator. During charging, lithium ions move from the cathode to the anode, and during discharging, they move back to the cathode to supply electrical energy to the connected device. Lithium-ion batteries are lightweight, have high energy density, and can be recharged many times. They are widely used in mobile phones, laptops, electric vehicles, power banks, and robotics projects. In robotics, lithium-ion batteries provide portable power to microcontrollers, sensors, and motors. Proper charging, voltage regulation, and protection against overcharging, over-discharging, and short circuits are essential for safe operation.

**Fig 2.14 Li-ion Cell**



## 2.2 FLOWCHART

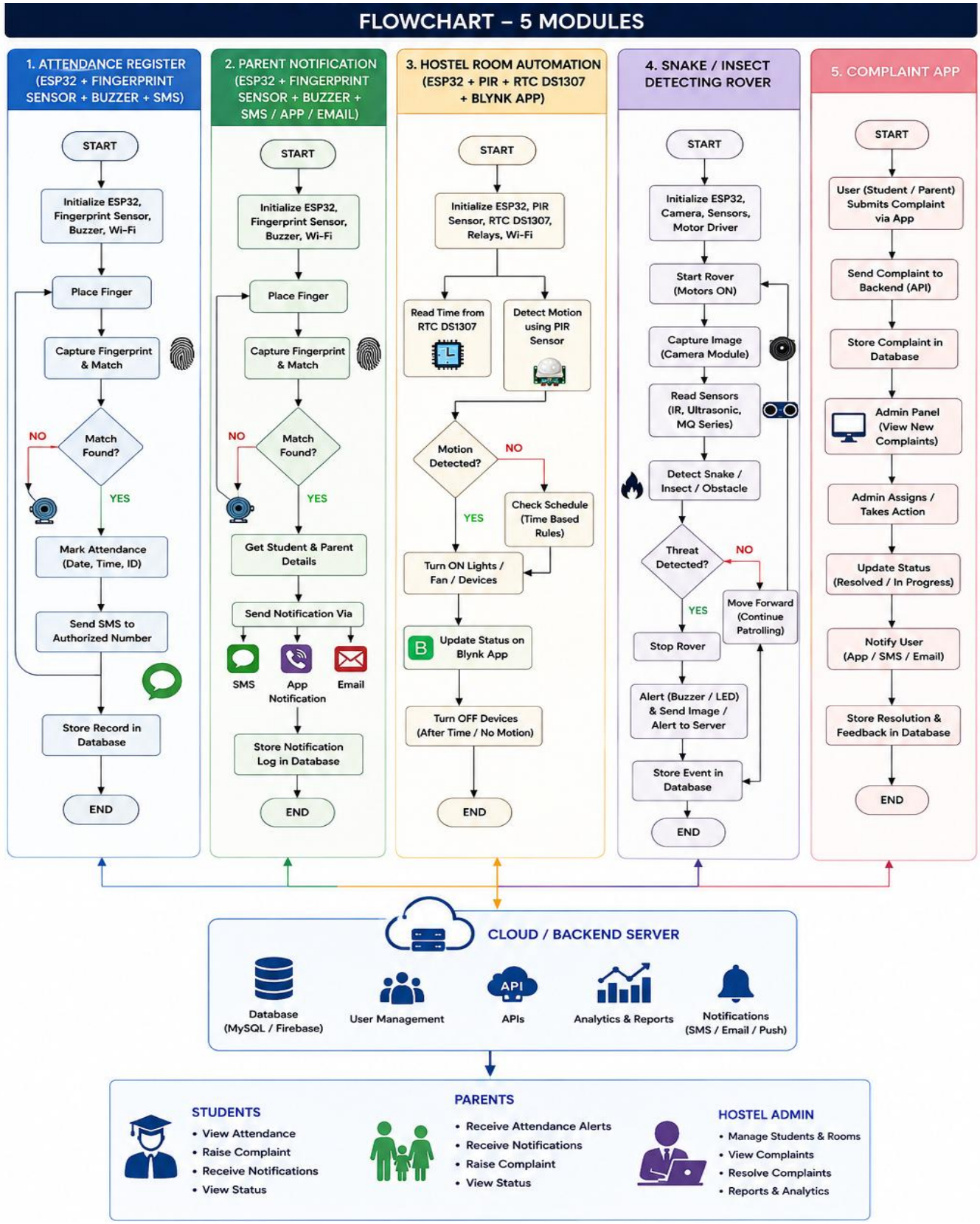
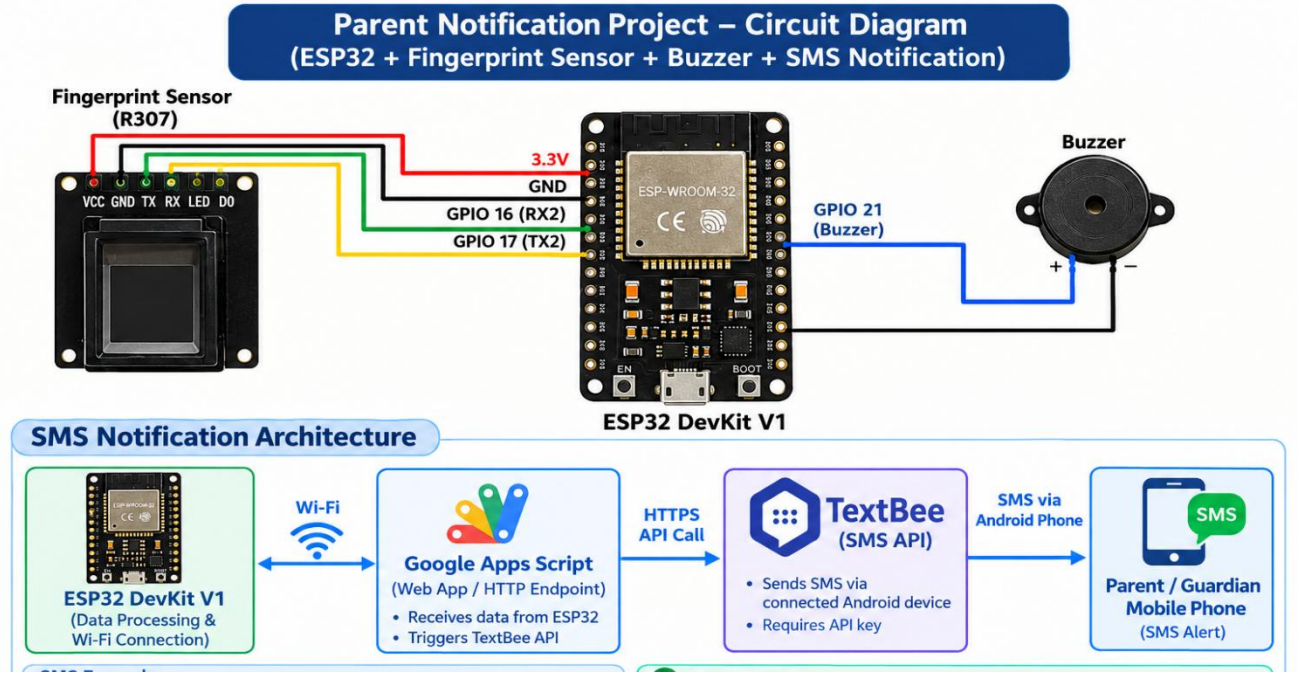



**Fig 2.15 Flowchart**

## 2.3 CIRCUIT DIAGRAM

### 2.3.1 Parent Notification




**2.16 Parent notification**

### 2.3.2 Attendance Register

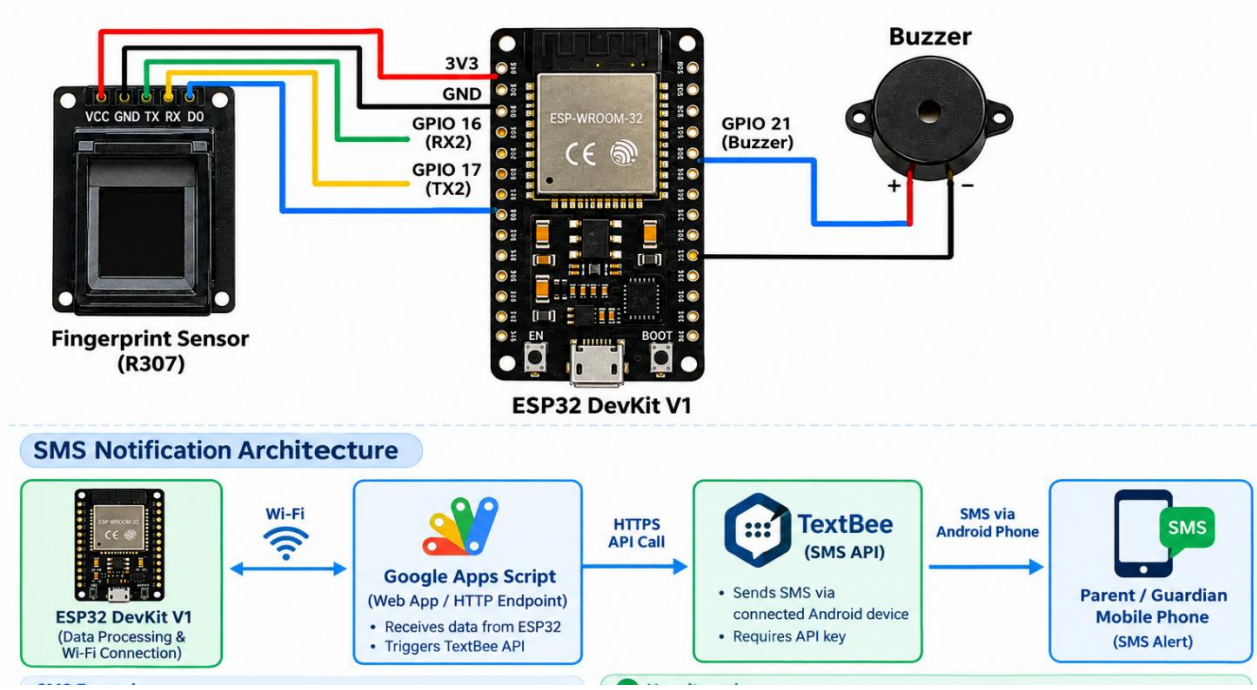
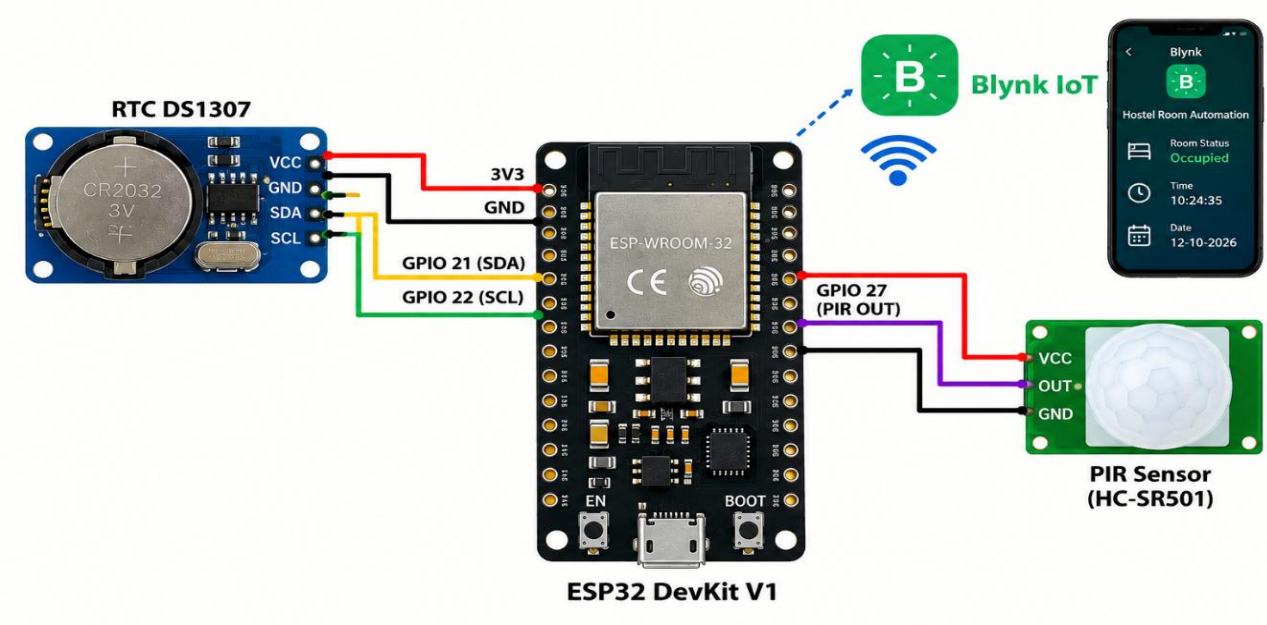


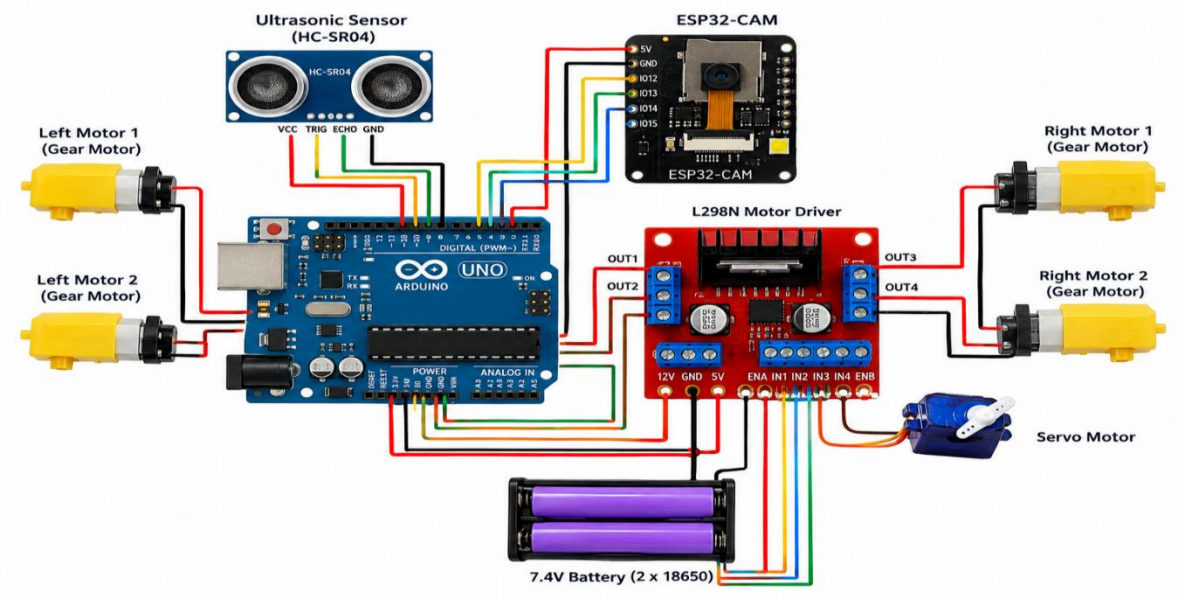

**Fig 2.17 Attendance Register**

## 2.3.3 Room Automation



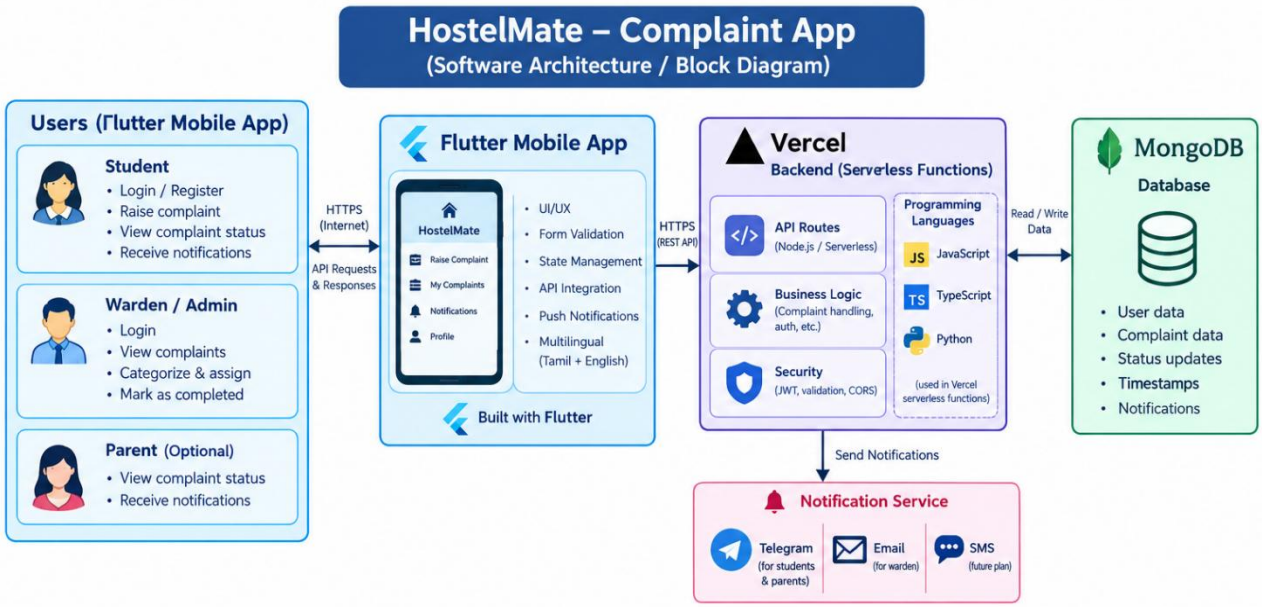

**2.18 Hostel Room Automation**

## 2.3.4: Snake/Insect Detecting Robot




**2.19 Snake/Insects Detecting Rover**

## 2.3.5 Complaint App




**2.20 Complaint App**

# CHAPTER 3

# OPERATION AND WORKING

## 3.1 PARENT NOTIFICATION

The Fingerprint-Based Parent Notification System is designed to improve hostel security and maintain communication between parents and hostel management. The system uses an ESP32 microcontroller and a fingerprint sensor to identify students during hostel entry and exit. Each registered fingerprint is linked to student details such as name, room number, and fingerprint ID.

When a student scans their fingerprint, the sensor verifies the identity and sends the information to the ESP32 for processing. The student's name, date, time, and entry/exit status are recorded in Google Sheets using Google Apps Script. This digital record reduces manual work and enables hostel administrators to track student movements efficiently.

An SMS notification feature is being implemented to send automatic alerts to parents registered mobile numbers whenever their children enter or leave the hostel. The messages are designed in both Tamil and English for better accessibility. The system is tested for fingerprint recognition, accurate record storage, date and time handling, and successful SMS delivery.

Overall, the system integrates biometric authentication, cloud-based data storage, and SMS communication to improve student safety, simplify hostel administration, and keep parents informed about their children's hostel movements.

## 3.2 ATTENDANCE REGISTER

The Fingerprint-Based Attendance Register System is developed using an ESP32 microcontroller and a fingerprint sensor to automate student attendance monitoring in the hostel. The system aims to improve attendance accuracy, reduce manual work, and simplify hostel administration. Each student's fingerprint is registered and linked to personal details such as name, room number, and block number.

The system records attendance daily between 3:00 PM and 6:00 PM. When a student scans their fingerprint, the sensor verifies the identity, and the ESP32 records the student's name, attendance status, date, and time in Google Sheets. Google Apps Script is used to process and manage attendance data automatically.

At the end of the attendance session, students who have scanned their fingerprints are marked as present, while those who have not scanned are identified as absent. The absentee list is prepared automatically and sent to the hostel warden through email. Parents also receive SMS notifications about their ward's absence, enabling timely communication.

Overall, the system combines biometric authentication, cloud-based data storage, email reporting, and SMS notifications to provide a reliable, efficient, and automated hostel attendance management solution.

## 3.3 HOSTEL ROOM AUTOMATION

The Hostel Room Automation System is developed using an ESP32 microcontroller, a Real-Time Clock (RTC) module, and a Passive Infrared (PIR) sensor to monitor hostel room activity during specified hours. The main objective is to improve hostel security and help wardens monitor room activity remotely.

The system operates during two predefined time intervals: 9:00 AM to 12:00 PM and 2:00 PM to 4:00 PM. The RTC module provides the current time and date, while the PIR sensor detects human movement within the room. The ESP32 processes the sensor signals and checks whether the detected movement occurs during the scheduled monitoring periods.

When movement is detected during the specified hours, the ESP32 sends room status information to the Blynk IoT platform through Wi-Fi. The warden can monitor room activity and receive real-time notifications through the Blynk application on a smartphone. This reduces the need for frequent manual room inspections and supports timely action when required.

The system is developed using the Arduino IDE, with the RTC, PIR sensor, ESP32, and Blynk platform integrated and tested for accurate time-based monitoring and notification delivery. Since the PIR sensor detects movement rather than identifying individuals, additional verification may be required when confirming a person's identity.

Overall, the system provides an IoT-based solution for scheduled hostel room monitoring, supporting remote supervision, improved security awareness, and efficient hostel management.

## 3.4 SNAKES / INSECTS DETECTING ROVER

The Hostel Room Automation System is designed using an ESP32 microcontroller, a Real-Time Clock (RTC) module, and a Passive Infrared (PIR) sensor to monitor hostel room activity during predefined hours. The main objective is to enhance hostel security, support remote supervision, and improve administrative efficiency.

The system operates between 9:00 AM and 12:00 PM, and 2:00 PM and 4:00 PM. The RTC module provides accurate date and time information, while the PIR sensor detects human movement inside the room. The ESP32 processes the sensor signals and

determines whether movement occurs during the scheduled monitoring periods.

When movement is detected during the specified hours, the ESP32 transmits the room status to the Blynk IoT platform through Wi-Fi. The warden can monitor room activity remotely and receive real-time notifications through the Blynk mobile application. This reduces the need for frequent manual inspections and enables timely action when necessary.

The system is programmed using the Arduino IDE and tested to verify sensor operation, time-based monitoring, Wi-Fi connectivity, and notification delivery. Although the PIR sensor detects movement, it cannot identify a specific person.

Overall, the system integrates ESP32, RTC, PIR, Wi-Fi, and Blynk IoT to provide an automated hostel room monitoring solution that improves security awareness, simplifies supervision, and supports efficient hostel management.

## 3.5 HOSTEL MATE COMPALAINT APP

The Hostel Mate Complaint Management Application is designed to simplify the submission, tracking, and resolution of hostel-related complaints. Its main objective is to improve communication between students and hostel management and ensure that maintenance issues are addressed efficiently. Students can report problems related to water supply, electrical faults, room maintenance, cleanliness, and other hostel facilities through a user-friendly mobile application.

The application allows students to view their profile details, including name, block number, and room number. When submitting a complaint, students can enter a title and description, and the system assigns a unique complaint ID for easy tracking. The complaint details, student information, submission date, and current status are stored in the database.

The application provides four complaint statuses: Submitted, Received, In Progress, and Resolved. Students can view their previous complaints, search using complaint IDs or keywords, and track status updates. The warden or administrator can access a dashboard to review complaints, identify the reported location, assign necessary actions, and update the status after resolving the issue.

The application is tested for complaint submission, database integration, complaint ID generation, search functionality, profile management, and status updates. It reduces manual paperwork, minimizes communication gaps, and improves accountability in hostel maintenance.

Overall, Hostel Mate provides a centralized digital platform for efficient complaint management, helping students report problems easily and enabling hostel administrators to monitor and resolve issues systematically.

# CHAPTER 4

# IMPLEMENTATION AND RESULTS

## 4.1 WOKWI SIMULATOR

Wokwi is an online electronics simulator used to design, simulate, and test microcontroller-based circuits without requiring physical hardware. In the AIoT-Driven Smart Hostel Management and Student Protection Ecosystem, Wokwi is used to simulate the ESP32 microcontroller and its interfacing with various electronic components. It supports embedded C/C++ programming using the Arduino framework and helps verify circuit connections and program logic before hardware implementation.

Wokwi enables the simulation of components such as sensors, LEDs, buzzers, LCD displays, and other supported modules. It helps test the hostel room automation, water-level monitoring, and home automation modules by observing sensor inputs and corresponding output responses. The serial monitor displays sensor readings, debugging messages, and system status information, making it easier to identify programming errors and incorrect pin configurations.

The simulator reduces initial hardware testing costs, minimizes wiring errors, and saves development time. It also helps demonstrate the working principles of individual modules during project reviews and presentations.

Overall, Wokwi provides a convenient platform for circuit simulation, program debugging, and preliminary testing, supporting the reliable development of the smart hostel management system before practical hardware implementation.

## 4.2 ARDUINO IDE

Arduino IDE (Integrated Development Environment) is a software tool used to write, compile, and upload embedded C/C++ programs to microcontrollers such as ESP32 and ESP8266. In the AIoT-Driven Smart Hostel Management and Student Protection Ecosystem, it is used to develop and test the programs required for various automation and safety modules.

Arduino IDE provides a simple code editor and supports libraries for interfacing with sensors, fingerprint modules, RTC modules, ultrasonic sensors, LCD displays, relays, and Wi-Fi communication. The appropriate board, communication port, and required libraries are selected before uploading the program to the microcontroller.

In this project, Arduino IDE is used to program fingerprint-based attendance and

parent notification systems, hostel room automation, water-level monitoring, home automation, and the snake-detecting rover. It processes sensor inputs, controls output devices, manages communication with IoT platforms, and supports data transmission and alert generation.

The Serial Monitor helps display sensor readings, system messages, and debugging information. The compilation process identifies programming errors, while repeated testing helps verify the correct operation of each module.

Overall, Arduino IDE plays an important role in the programming, debugging, testing, and implementation of the smart hostel project, providing a common development platform for its microcontroller-based systems.

## 4.3 GOOGLE APPS SCRIPT

Google Apps Script is a cloud-based scripting platform that uses JavaScript to automate tasks and integrate Google services such as Google Sheets and Gmail. In the AIoT-Driven Smart Hostel Management and Student Protection Ecosystem, it is used to manage student attendance records, hostel entry/exit logs, and parent notifications.

In the Parent Notification System, the ESP32 sends student fingerprint ID and entry/exit information to the Google Apps Script web application after successful fingerprint verification. The script retrieves the corresponding student details and stores the student's name, date, time, and movement status in Google Sheets. It also integrates with an SMS service to send notifications to registered parents in Tamil and English.

In the Fingerprint-Based Attendance Register System, Google Apps Script records attendance between 3:00 PM and 6:00 PM and helps identify students who have not marked their attendance. The absentee report can be sent to the hostel warden through email, while parents can receive SMS alerts about their ward's absence.

Google Sheets acts as a centralized database, while functions such as doGet() and doPost() support communication between the ESP32 and cloud services. The script also helps maintain accurate date and time records using the Indian Standard Time zone.
Overall, Google Apps Script automates data processing, attendance management, and notification delivery, reducing manual work and improving communication between hostel administrators and parents.

## 4.4 BLYNK IoT PLATFORM

The Blynk IoT platform is used in the AIoT-Driven Smart Hostel Management and Student Protection Ecosystem to monitor connected hardware modules remotely through the internet. It provides a user-friendly mobile interface for displaying sensor readings, checking device status, and receiving real-time notifications. The ESP32 communicates with the Blynk cloud platform through Wi-Fi to transmit relevant system information.

In the hostel room automation module, Blynk displays room monitoring information and provides notifications when movement is detected during the predefined monitoring periods. The ESP32 processes inputs from the PIR sensor and RTC module and updates the room status on the Blynk dashboard.

The Blynk dashboard can be customized using widgets such as indicators, gauges, buttons, and notification features. During development, the device configuration, data streams, and dashboard settings are tested to ensure proper communication and accurate updates.

Overall, Blynk provides remote monitoring and notification capabilities that improve system accessibility, support timely responses, and reduce the need for continuous manual supervision in hostel management.

## 4.5 Flutter App

Flutter is an open-source framework developed by Google for building cross-platform mobile applications using a single codebase. In the AIoT-Driven Smart Hostel Management and Student Protection Ecosystem, Flutter is used to develop the Hostel Mate Complaint Management Application, which simplifies communication between students and hostel management.

The application provides a user-friendly interface for students to view their profiles and submit complaints related to electrical faults, water supply, room maintenance, cleanliness, and other hostel facilities. Student details, including name, block number, and room number, help administrators identify the location of reported problems.

When a complaint is submitted, the application assigns a unique complaint ID and stores the complaint details in the connected database. Students can search for complaints and track their progress through four statuses: Submitted, Received, In Progress, and Resolved. The administrator dashboard enables the warden to review complaints, update their status, and monitor maintenance activities.

Flutter supports the development of application screens, forms, buttons, navigation menus, and status indicators. Its widget-based architecture helps create a responsive and consistent user interface. The application is tested to verify complaint submission, data retrieval, search functionality, and status updates.

Overall, Flutter provides an effective platform for developing the Hostel Mate application, reducing paperwork, improving transparency, and supporting efficient complaint management in the hostel.

## 4.6 TECHNOLOGIES STACK

## FRONTEND:

The frontend of the Hostel Mate Complaint Management Application is developed using Flutter and Dart to provide a user-friendly and responsive interface for students and hostel administrators. Flutter supports cross-platform application development, while Dart is used to implement the application logic, interface components, and user interactions.

Provider is used for state management to update the interface when complaint details or statuses change. Go Router enables navigation between screens such as login, student profile, complaint submission, complaint tracking, and the administrator dashboard. Dio and HTTP support communication with backend services through API requests, while Flutter Secure Storage helps securely store sensitive information such as authentication tokens.

Google Fonts improves text presentation and readability. The Flowchart package supports the visualization of complaint statistics, while Lottie animations enhance the user experience and Shimmer provides loading placeholders while data is being retrieved.

These technologies support complaint registration, complaint ID-based searching, status tracking, profile viewing, and administrator complaint management. Overall, the frontend technology stack improves application usability, navigation, data presentation, and communication between students and hostel management.

### BACKEND:

The backend of the Hostel Mate Complaint Management Application is developed using Node.js and Express.js to process application requests, manage data, and handle complaint-related operations. Node.js provides the server-side runtime environment, while Express.js simplifies the development of RESTful APIs that connect the Flutter mobile application with backend services.

The backend follows the Model-View-Controller (MVC) architecture, which organizes the application into separate components for data management, user interface interaction, and request processing. This structure improves code organization, maintainability, and scalability.

RESTful APIs support complaint submission, complaint ID-based searching, student profile retrieval, complaint tracking, and status updates. The system manages complaint statuses such as Submitted, Received, In Progress, and Resolved, enabling administrators to monitor maintenance requests and students to track their progress.

The Groq SDK can provide AI-based features, including automatic complaint categorization and analysis of frequently reported issues. Vercel is used to deploy and host the backend API, making it accessible to the mobile application through the internet.

Overall, Node.js, Express.js, MVC architecture, Groq SDK, and Vercel provide the backend foundation for efficient complaint processing, AI-assisted analysis, and reliable communication between students and hostel administrators.

**DATABASE:**

The database and cloud services of the Hostel Mate Complaint Management Application are used to store student information, manage complaint records, provide secure authentication, and handle complaint-related images and attachments. These services ensure organized data management and convenient access for students and hostel administrators.

MongoDB is used to store student profiles, complaint IDs, descriptions, room numbers, block details, submission dates, and complaint statuses. Mongoose ODM helps define data models, validate information, and perform database operations between the Node.js backend and MongoDB.

The Firebase Admin SDK can support push notifications through Firebase Cloud Messaging. Google Auth Library enables Google authentication when configured, while JSON Web Tokens (JWT) help manage authenticated sessions and protect authorized API endpoints.

Cloudinary is used to store complaint-related images and attachments, such as photographs of damaged electrical equipment, leaking pipes, and other maintenance issues. The backend stores the corresponding media references with complaint records, allowing administrators to review supporting images.

Overall, MongoDB, Mongoose, Firebase Admin SDK, Google Auth Library, JWT, and Cloudinary provide essential database, authentication, notification, and media storage services. These technologies support secure data management, efficient complaint tracking, and improved hostel maintenance operations.

## 4.7 OBSERVATION AND OUTPUT

A IoT-Driven Smart Hostel Management and Student Protection Ecosystem was observed and evaluated to understand the performance of its individual modules and their overall functionality. The system integrates fingerprint-based attendance, smart hostel monitoring, an ESP32-CAM-based safety robot, parent notification, and a hostel complaint management application. Each module is designed to improve hostel security, monitoring, communication, and administrative efficiency.

During testing, the fingerprint sensor was used to identify registered students and support entry and exit recording. The PIR sensor detected movement, while the LDR sensor monitored light conditions for automatic lighting control. The RTC module provided time-based monitoring, and the relay controlled the lighting according to the programmed conditions. These functions helped reduce manual intervention and unnecessary power consumption.

The ESP32-CAM-based safety robot was designed to capture images of the surrounding environment and support the identification of potential hazards. When integrated with a suitable AI detection model, the robot can identify suspected snakes or other dangerous objects and trigger an alert. The parent notification module was designed to send SMS alerts regarding important student-related events through the configured SMS service. The complaint management application, deployed using the Vercel platform, enabled students to submit complaints and track their status through the stages of Pending, In Progress, and Resolved.

The observed outputs demonstrate that integrating embedded systems, sensors, IoT connectivity, and web-based applications can improve hostel management. The system provides a more organized approach to attendance monitoring, environmental control, hazard alerts, parent communication, and complaint handling.

The expected outputs of the proposed system are listed below:

* Fingerprint Attendance: Registered students can be identified, and their entry or exit events can be recorded.

* Smart Hostel Monitoring: PIR and LDR sensors support movement detection and automatic lighting control.

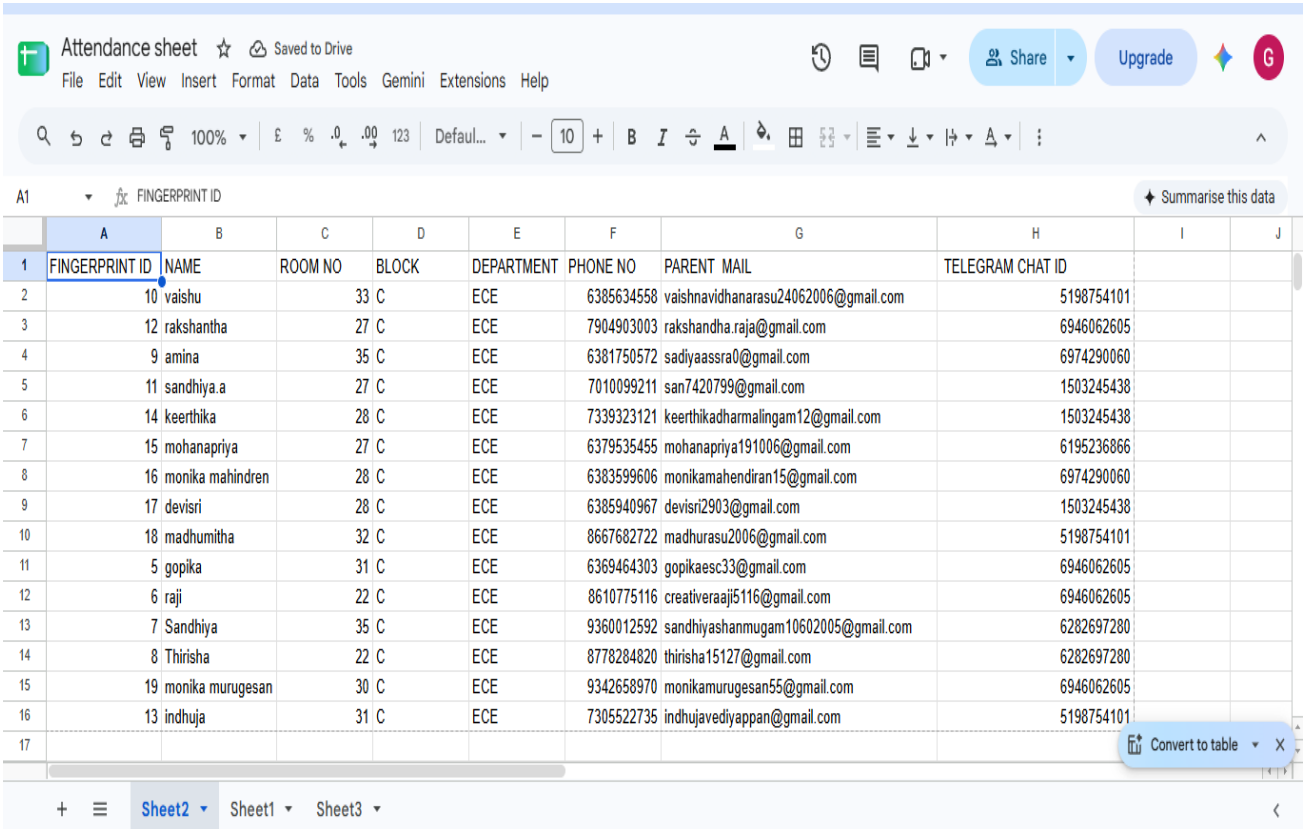
* Safety Robot: The ESP32-CAM captures images to support hazard detection and alert generation.

* Parent Notification: SMS alerts can be sent to registered parent numbers when the configured events occur.

* Complaint Management: Students can submit complaints online, and administrators can update their status.

* IoT Monitoring: Relevant system information can be viewed remotely through the configured IoT platform.

Overall, the proposed ecosystem provides a centralized and systematic approach to hostel management. It aims to improve student safety, simplify administrative tasks, reduce manual effort, and provide better communication between students, parents, and hostel administrators. The final performance depends on the successful implementation and testing of each module.

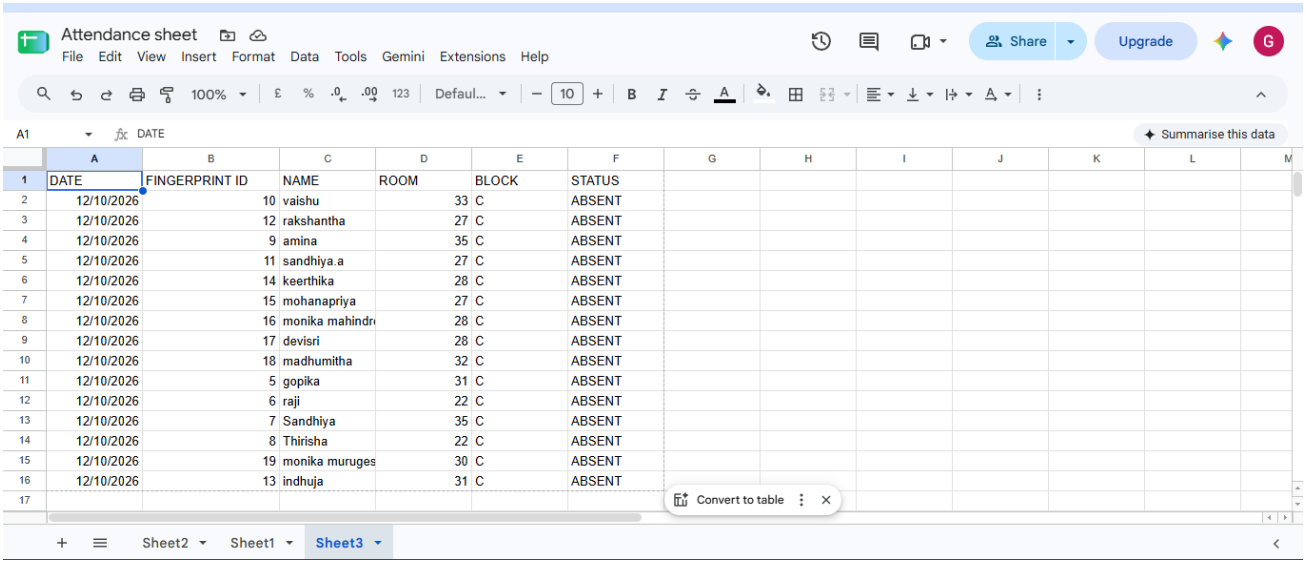



| FINGERPRINT ID | NAME | ROOM NO | BLOCK | DEPARTMENT | PHONE NO | PARENT MAIL | TELEGRAM CHAT ID |
|---|---|---|---|---|---|---|---|
| 10 | vaishu | 33 | C | ECE | 6385634558 | vaishnavidhanarasu24062006@gmail.com | 5198754101 |
| 12 | rakshantha | 27 | C | ECE | 7904903003 | rakshandha.raja@gmail.com | 6946062605 |
| 9 | amina | 35 | C | ECE | 6381750572 | sadiyaassra0@gmail.com | 6974290060 |
| 11 | sandhiya.a | 27 | C | ECE | 7010099211 | san7420799@gmail.com | 1503245438 |
| 14 | keerthika | 28 | C | ECE | 7339323121 | keerthikadharmalingam12@gmail.com | 1503245438 |
| 15 | mohanapriya | 27 | C | ECE | 6379535455 | mohanapriya191006@gmail.com | 6195236866 |
| 16 | monika mahindren | 28 | C | ECE | 6383599606 | monikamahendiran15@gmail.com | 6974290060 |
| 17 | devisri | 28 | C | ECE | 6385940967 | devisri2903@gmail.com | 1503245438 |
| 18 | madhumitha | 32 | C | ECE | 8667682722 | madhurasu2006@gmail.com | 5198754101 |
| 5 | gopika | 31 | C | ECE | 6369464303 | gopikaesc33@gmail.com | 6946062605 |
| 6 | raji | 22 | C | ECE | 8610775116 | creativeraaji5116@gmail.com | 6946062605 |
| 7 | Sandhiya | 35 | C | ECE | 9360012592 | sandhiyashanmugam10602005@gmail.com | 6282697280 |
| 8 | Thirisha | 22 | C | ECE | 8778284820 | thirisha15127@gmail.com | 6282697280 |
| 19 | monika murugesan | 30 | C | ECE | 9342658970 | monikamurugesan55@gmail.com | 6946062605 |
| 13 | indhuja | 31 | C | ECE | 7305522735 | indhujavediyappan@gmail.com | 5198754101 |

**Fig 2.21 Google sheet output**




**Fig 2.22 Exit and Entry Notification**




| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | DATE | FINGERPRINT ID | NAME | ROOM | BLOCK | STATUS |
| 2 | 12/10/2026 | 10 | vaishu | 33 | C | ABSENT |
| 3 | 12/10/2026 | 12 | rakshantha | 27 | C | ABSENT |
| 4 | 12/10/2026 | 9 | amina | 35 | C | ABSENT |
| 5 | 12/10/2026 | 11 | sandhiya.a | 27 | C | ABSENT |
| 6 | 12/10/2026 | 14 | keerthika | 28 | C | ABSENT |
| 7 | 12/10/2026 | 15 | mohanapriya | 27 | C | ABSENT |
| 8 | 12/10/2026 | 16 | monika mahindr | 28 | C | ABSENT |
| 9 | 12/10/2026 | 17 | devisri | 28 | C | ABSENT |
| 10 | 12/10/2026 | 18 | madhumitha | 32 | C | ABSENT |
| 11 | 12/10/2026 | 5 | gopika | 31 | C | ABSENT |
| 12 | 12/10/2026 | 6 | raji | 22 | C | ABSENT |
| 13 | 12/10/2026 | 7 | Sandhiya | 35 | C | ABSENT |
| 14 | 12/10/2026 | 8 | Thirisha | 22 | C | ABSENT |
| 15 | 12/10/2026 | 19 | monika muruges | 30 | C | ABSENT |
| 16 | 12/10/2026 | 13 | indhuja | 31 | C | ABSENT |
| 17 | | | | | | |

**Fig 2.23 Attendance Register  Google Sheet Output**

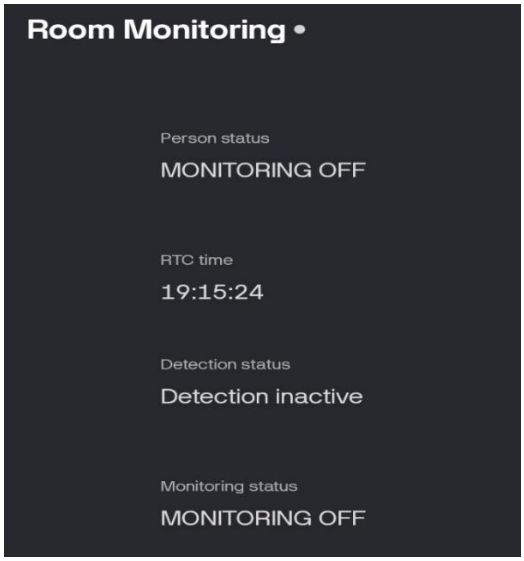



**Fig 2.24 :Room Automation output**

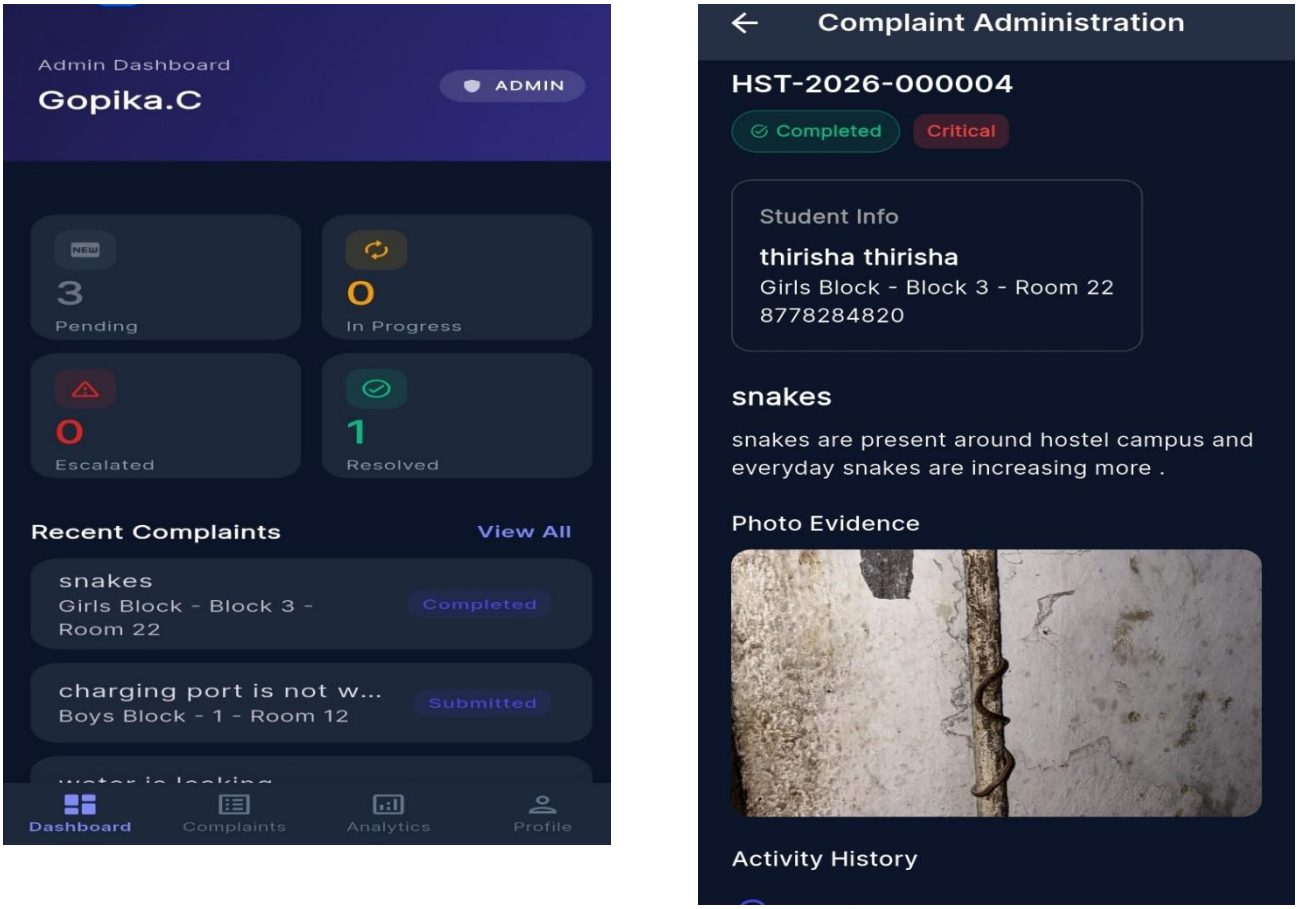



**Fig2.25 :Complaint app output**

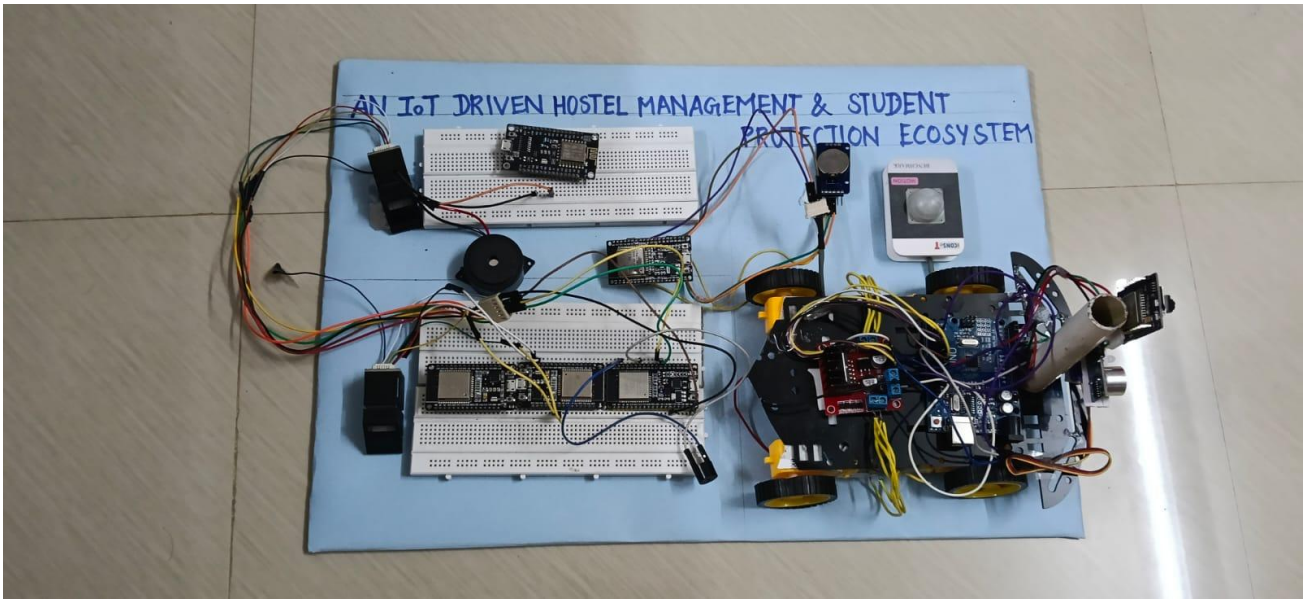



**Fig 2.26: Hardware setup image**

## 4.8 RESULTS AND DISCUSSION

A IoT-Driven Smart Hostel Management and Student Protection Ecosystem was tested to evaluate the functionality of its individual modules. The fingerprint-based attendance system was designed to identify registered students and record attendance and entry/exit activities, reducing manual record maintenance.

The hostel room monitoring module was tested using PIR and RTC sensors to detect movement during predefined hours. The LDR-based lighting control system was evaluated for automatic light operation under different lighting conditions. The water-level and home automation modules were also tested to verify sensor responses and the control of connected devices.

The ESP32-CAM-based safety robot was evaluated for potential hazard detection and alert generation. The parent notification system was tested for recording student movements and sending SMS alerts to registered parents. The Hostel Mate application was assessed for complaint submission, complaint ID generation, status tracking, and administrator updates.

Overall, the proposed system integrates automation, monitoring, notifications, and digital complaint management to support efficient hostel administration. It aims to reduce manual work, improve communication between students, parents, and wardens, and enhance hostel safety. Further testing under real hostel conditions is necessary to evaluate the system's long-term reliability and performance.

# CHAPTER 5

# CONCLUSION

## 5.1 Conclusion

The IoT-Driven Smart Hostel Management and Student Protection Ecosystem was successfully designed and developed to provide a smart, secure, and efficient solution for modern hostel management. The system integrates ESP32, fingerprint-based attendance, PIR and LDR sensors, RTC, relay control, an ESP32-CAM-based safety robot, IoT monitoring, SMS-based parent notifications, and a hostel complaint management application

Overall, the proposed system enhances student security, attendance management, energy conservation, communication, and complaint handling. It offers a cost-effective, scalable, and user-friendly approach to smart hostel management. In the future, the system can be further enhanced by integrating advanced AI algorithms, cloud-based data management, and additional automation features to support the development of safer and smarter hostel environments.

## 5.2 FUTURE ENHANCEMENT

The IoT-Driven Smart Hostel Management and Student Protection Ecosystem can be enhanced with advanced technologies to improve security, automation, and efficiency. AI-based face recognition can be integrated with fingerprint authentication to provide additional security during student entry and exit. Cloud-based storage can be implemented to maintain attendance records and enable remote access for hostel administrators.

Additional sensors, such as temperature, humidity, smoke, gas, and fire sensors, can be incorporated to monitor hostel conditions and generate emergency alerts. Automatic control of lights, fans, and other electrical appliances can be improved based on occupancy and environmental conditions to reduce energy consumption. The ESP32-CAM-based safety robot can be enhanced using AI and YOLO-based object detection to identify snakes and other potential hazards more accurately.

Autonomous navigation and live video streaming can support remote monitoring of hostel surroundings. The parent notification system can be expanded to include WhatsApp, email, and mobile application alerts alongside SMS notifications. The complaint management application can be improved with priority-based classification, image attachments, automatic assignment to maintenance staff, and real-time status updates. These enhancements can make the proposed system more intelligent, secure, scalable, and user-friendly, supporting the development of safer and smarter hostel environments in the future.

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